CERTIFICATION OF CONTROL SYSTEM SIGNAL CONDITIONING

Aero Metric Reference: UAV-IMU-LPF-04

Operational Domain: Flight Guidance, Navigation, and Control Core

Subsystem Classification: Digital Attitude Signal Restoration

1. Technical Framework

High-frequency mechanical oscillations caused by drone motor propulsion subsystems corrupt attitude estimation profiles. This document outlines the performance analysis of a discrete first-order lag recursive digital filter used to restore structural sensor telemetry.

2. Algorithmic Formulation

The sensor signal processing core evaluates the recursive tracking equation:

$$Y(k) = \alpha X(k) + (1 - \alpha)Y(k - 1)$$

Where X represents the raw sensor input vector, Y denotes the updated filtered attitude estimation, and α defines the design smoothing coefficient selected via target structural frequency limitations.

3. Signal Mitigation Analysis

- High-Frequency Attenuation: Testing verified strong mitigation of structural vibration profiles spanning the 80 Hz to 120 Hz bands.

- Variance Tracking Reduction: Total wideband signal variance dropped significantly, validating smoother data feeds for downstream processing.

- Phase Shift Properties: The calculated smoothing factor introduces a small, manageable phase delay that remains well within the stability margins of the primary flight control system.

4. Avionics System Integration Guidance

- Coefficient Optimization:*Set α to 0.08 for standard operations. Lower the value to 0.05 during high-wind operations to better handle increased aerodynamic body vibrations.

- Rate Tuning: The processing loop must execute deterministically at a fixed frequency matching the IMU hardware data update cycles.